Willian Pacheco Silva

Tech Lead | Software Engineering in Distributed Systems, AWS and DevOps

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SUMMARY

Technology professional working in solution architecture and technical leadership. I work across the full development cycle: structuring scalable systems, defining standards, guiding critical decisions and ensuring quality at every stage. Recent experience includes leading teams, designing enterprise solutions, integrating platforms and optimizing processes. Computer Engineer with a master's degree from UFES (Electrical Engineering program, focused on computing, AI and machine learning), with applied research and results published at IEEE.

EXPERIENCE

Loomis Desenvolvimento de Softwares Ltda. Jan 2026 - Present

Founder and owner

- Software development and architecture company: design, construction and maintenance of custom software.

2Solve Inteligência Tecnológica Mar 2024 - Present

Tech Lead (Jan 2026 - Present)

- Technical leadership of the software engineering team: critical architecture decisions, engineering standards, code quality and platform evolution.
- Hands-on work in development and code review, system architecture, cloud infrastructure and database decisions (modeling, performance, availability and reliability).

Mid-level Backend Developer II (Nov 2025 - Jan 2026)

- Technical leadership of strategic projects, with training in Portuguese and English, one-on-one follow-up and code reviews.
- Architecture and maintenance of the AWS infrastructure: Linux servers on EC2, microservices on ECS Fargate, pipelines, automation and provisioning with Terraform.

Backend Developer Level 2 (Mar 2024 - Nov 2025)

- Back-end development in Python, with Docker images, deployment on AWS EC2 instances and database integration.

Fundação Espírito-santense de Tecnologia (FEST) Jul 2023 - Jun 2026

Researcher (undergraduate research scholarship and research project alongside the master's)

- Project "Optical Fiber Profiler for FPSO Tanks" (since July 1, 2023): analysis of the profiler data to relate sensor readings to water and oil levels.

Superprof Jun 2025 - Present

Private tutor (Super Professor level)

- Personalized Mathematics, Programming and Music lessons for students of different levels and goals.

Quantum Tecnologia Jun 2023 - Jan 2024

AI Development Intern (Full Cycle)

- Exploratory analysis of AI technologies and time series forecasting; use of LLMs for text and audio; tools and chatbots with OpenAI.
- Python with clean code and object-oriented design; Docker, CI/CD, APIs with Swagger, documentation with PlantUML, OpenProject, Git and GitLab.

Controle Eletrônica
Jan 2018 - Dec 2018

Full Stack Developer (web and desktop applications)

- Implementation and maintenance of the company's website and ERP; microcontroller programming.
- Pascal, C, C++ and Python; MySQL and PostgreSQL; HTML, CSS, JavaScript and Bootstrap; Docker.

EDUCATION

Master's in Electrical Engineering, UFES
2024 - 2026

Electrical Engineering program, focused on computing, AI and machine learning. Dissertation (defended on June 23, 2026): Sensor fusion based on FBG sensors for robust liquid level estimation in noisy environments. Advisor: Prof. Dr. Helder Roberto de Oliveira Rocha. Co-advisor: Profa. Dra. Maria José Pontes.

Bachelor's in Computer Engineering, UFES
2019 - 2023

Thesis: Analysis of sensor fusion in convolutional neural networks for liquid level prediction with FBG optical sensors. Advisor: Prof. Dr. Guilherme Fernandes de Souza Miguel.

Technical Course in Systems Development, SENAI-ES
2017 - 2018

Recognitions in 2018: highest overall grade in the class and 100% attendance.

High school, SESI-ES
2016 - 2018

PUBLICATIONS

FIOROTTI, R.; YAHYAOUI, I.; **SILVA, W. P.**; MACHADO, A. A. P.; FARDIN, J. F.; SEGATTO, M. E. V.; ROCHA, H. R. O. Towards smart EVs energy scheduling for a logistic distribution center: cost, emissions and makespan optimizing. *Energy Conversion and Management: X*, v. 31, art. 102017, 2026. DOI 10.1016/j.ecmx.2026.102017.

SILVA, W. P.; LEAL JUNIOR, A. G.; SEGATTO, M. E. V.; PONTES, M. J.; ROCHA, H. R. O. Robustness of Liquid Neural Networks to Noise: A Comparative Study with Convolutional Neural Networks. In: SBMO/IEEE MTT-S International Microwave and Optoelectronics Conference (IMOC), 2025, Campina Grande. p. 429-433. DOI 10.1109/imoc65414.2025.11365756.

SILVA, W. P.; ROCHA, H. R. O.; FABER, M. J.; KLEINE DETERS, J.; BERGSMA, E.; SILVA, J. A. L. Fine-Tuning Large Language Models to Detect Critical Scenarios in Dementia Patient Monitoring and to Reduce Bandwidth Usage. In: IEEE International Conference on Industry Applications (INDUSCON), 2025, São Sebastião. p. 2214-2219. DOI 10.1109/induscon66435.2025.11241782.

SILVA, W. P.; FIOROTTI, R.; ROCHA, H. R. O.; YAHYAOUI, I. Minimization of Energy Expenditure and Charging Cycles in Distribution Systems With Electric Vehicles Using MOGWO. In: International Renewable Energy Congress (IREC), 2025, Hammamet. p. 1-6. DOI 10.1109/irec64614.2025.10926778.

SKILLS

Architecture and leadership: software architecture, distributed systems, technical leadership, mentoring.

Cloud and DevOps: AWS (EC2, ECS Fargate), Terraform, Docker, Linux, CI/CD, Git and GitLab.

AI and Machine Learning: TensorFlow, PyTorch, Keras, Optuna, neural networks (MLP, CNN, LNN), Random Forest, Transformers, LLMs and OpenAI, time series, metaheuristic optimization (MOGWO).

Development: Python, JavaScript, TypeScript, React, Next.js, Node.js, PHP, Java, C, C++, C#, .NET, HTML, CSS, Sass, Bootstrap, jQuery, Flask.

Databases: PostgreSQL, MySQL, SQL Server.

Languages: Portuguese (native), English (full professional), Spanish (elementary).